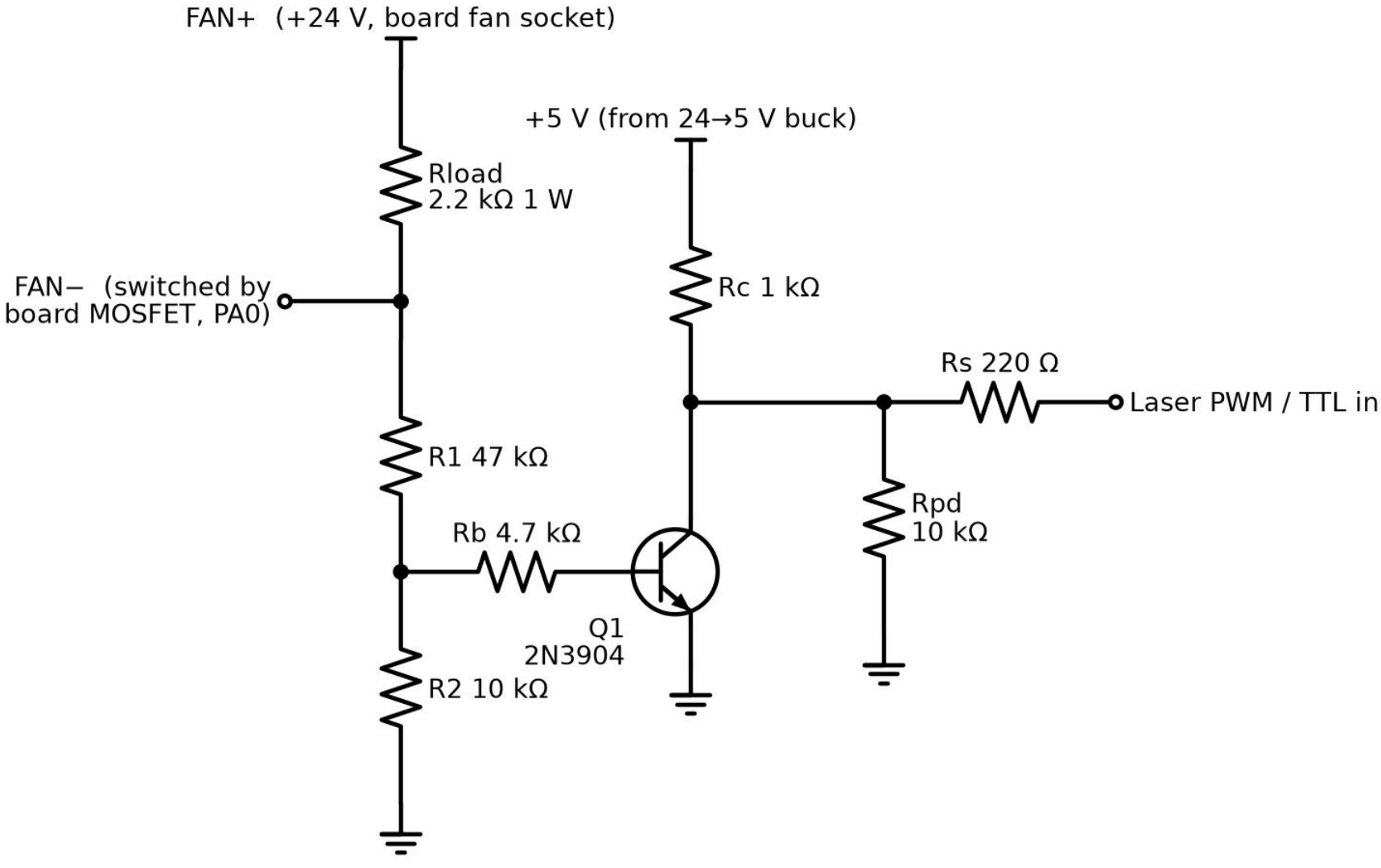


Fan output ON → FAN− ≈ 0 V → Q1 off → OUT ≈ 4.5 V (laser ON)
Fan output OFF → FAN− ≈ 24 V → Q1 on → OUT ≈ 0 V (laser OFF)
No 5 V / board off → Rpd holds OUT low
All grounds common: PSU DC−, buck GND, laser GND. Optional 100 nF across +5 V–GND.